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ORIGINAL ARTICLE

Enhancing part-based gait recognition via ensemble learning and feature fusion

Büşranur Yaprak[1] · Eyüp Gedikli[2]

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Abstract

Gait, a behavior-based biometric feature, has gained increasing popularity in human identification, particularly in surveillance systems, due to its ability to function without physical contact or explicit consent. Traditional silhouette-based methods have demonstrated that different body parts exhibit distinct movement patterns during walking, thereby enhancing recognition accuracy. In this study, we propose an improved part-based gait recognition approach by leveraging ensemble learning on local body regions. The Gait Energy Image (GEI) is segmented into five horizontal parts, and ensemble learning is applied to the convolutional neural network (CNN) responsible for their processing. A separate MetaModel is trained for each body part to integrate the part-based features obtained from ensemble learning and synthesize the most discriminative ones. Additionally, a part-removal process is introduced to mitigate the effects of appearance-based variations by analyzing absolute differences between images with and without variations. The aggregated most distinctive features contribute to robust recognition. We evaluate our proposed approach on the CASIA-B, CASIA-C, and Outdoor-Gait datasets, and experimental results indicate that ensemble learning significantly enhances part-based gait recognition performance under various appearance variations, outperforming several state-of-the-art methods. The datasets and source code are available at https://github.com/busrakckugurlu/Enhancing-Part-based-Gait-Recognition-via-Ensemble-Learning-and-Feature-Fusion/tree/main.

Keywords Ensemble learning · Gait recognition · Part-based gait recognition · Biometrics

1 Introduction

Gait recognition is a biometric technology used to identify individuals based on their unique body shapes and walking styles. It enables the recognition of persons from a considerable distance or at low resolution, which is often not possible with other biometric methods such as face, fingerprint, or palm print recognition. Furthermore, gait recognition does not require active cooperation from the subject, making it particularly advantageous for video surveillance applications. However, the performance of gait recognition is affected by various variations, including cross-view variations, carrying or clothing conditions, or walking speed changes. One of the key challenges in gait recognition is to minimize the impact of these variations and extract invariant gait features. Various methods have been proposed to extract gait features. For example, a Gait Energy Image (GEI) [1] is generated by averaging the aligned Q2silhouettes of gait sequences into a single image. Similarly, Chrono Gait Image (CGI) [2], Gait Entropy Image (GEnI) [3], Gait Flow Image (GFI) [16], and Period Energy Image (PEI) [4] also aggregate the gait cycle into one image to represent gait. Recently, numerous approaches have utilized convolutional neural networks (CNNs) to generate gait feature representations, leading to significant performance improvements. These methods can be categorized into global and local feature-based approaches [5]. Global feature-based approaches extract features from the whole silhouette. For instance, Wu et al. [8] presented a CNN-LB method that automatically learns distinctive gait features

✉ Büşranur Yaprak
busra.kucukugurlu@gumushane.edu.tr

Eyüp Gedikli
gediklie@ktu.edu.tr

1 Department of Software Engineering, Gümüşhane University, 29100 Gümüşhane, Turkey

2 Department of Software Engineering, Karadeniz Technical University, 61080 Trabzon, Turkey

from GEI pairs. Chao et al. [6] extracted the frame-level global features from unordered silhouette images. Jimenez et al. [9] introduced a multi-modal network that fuses four different global modalities, including silhouettes. In contrast, local feature-based approaches focus on extracting features from specific regions of the silhouettes and then combining them for improved recognition. For instance, Zhang et al. [7] divide the silhouettes into four horizontal parts and fuse the features extracted from each part using separate CNNs. Fan et al. [10] introduced a temporal part-based framework that includes a focal convolution layer. Lin et al. [5] designed a 3D CNN framework to extract both global and local features for discriminative gait representations. Based on the performance of the aforementioned part-based methods, this paper improves the overall performance of a simple part-based CNN architecture by applying ensemble learning, which enhances the robustness of features extracted by CNN models for each body part. Several reasons can be listed as to why ensemble learning frequently enhances predictive performance [44]:

- When only a small dataset is available, a model may learn patterns that fit the training data perfectly but fail to generalize to unseen data. Averaging predictions from multiple models reduces the risk of choosing an incorrect prediction, thus improving overall prediction performance.
- Single models that rely on local search algorithms may get trapped in local minima. However, by combining multiple learners, ensemble learning reduces the risk of obtaining a local minimum.
- A correct prediction may be outside the search space of a single model. By aggregating multiple models, the search space can be expanded or varied, leading to a more suitable search area for the data and increasing the probability of reaching the correct prediction.

Ensemble learning techniques have demonstrated remarkable success in various domains, particularly in human activity recognition, person re-identification, and action recognition, where their ability to improve prediction accuracy through model diversity has been well-established. For instance, in the field of human activity recognition, Gupta and Semwal [45] introduced an ensemble learning approach that applies different classification algorithms such as Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Extreme Learning Machine (ELM), and Principal Component Analysis (PCA) for multi-activity gait classification based on the fact that the combination of different classifiers provides average performance that avoids overfitting and is less dependent on hyper-parameters. In another study [46], it has been proven that the proposed ensemble learning hybrid model based on CNNs outperforms each of the other hybrid models. In person re-identification, Ye et al. [47] proposed a collaborative ensemble learning strategy that improves the performance of a two-stream CNN by facilitating knowledge transfer among different classifiers with fewer hyperparameters. In a separate study [48], a multi-branch architecture with self-ensemble learning was introduced, which utilizes self-ensemble prediction to increase the quality of mean teacher-based soft labels. In a study for three-dimensional (3D) action recognition conducted by Liang et al. [49], one stage of the proposed three-stream CNN incorporates a multitask and ensemble learning network to enhance its generalization ability. Xu et al. [50] proposed an ensemble neural network for skeleton-based action recognition, consisting of four one-dimensional (1D) CNN subnets with residual structures to extract diverse and complementary features. Then, they fused four subnets into a single ensemble network, achieving remarkable performance. While ensemble learning has been proven to yield successful results in areas such as human activity recognition, person re-identification, and action recognition, its application in gait recognition has remained relatively underexplored and limited. For instance, in a study proposed by Wang and Yan [22], a new gait feature is extracted based on the distance between two gait regions, then a set of view-dependent Hidden Markov Model (HMM) learners are trained separately according to different views of the same individual. Finally, a set of view-dependent learners are ensembled to classify gaits. In another study [21], random sampling with a replacement strategy is used to generate a series of training sets from GEIs, then primary CNNs with different hyperparameters are trained, and finally, a secondary classifier is used to ensemble CNN models. However, these studies have certain methodological and experimental limitations. In particular, the absence of gallery and probe dataset partitioning makes it difficult to compare their results with state-of-the-art studies, which is essential for demonstrating the effectiveness of ensemble learning. In this study, we employ two distinct ensemble learning approaches—k-fold cross-validation and bootstrap aggregating—to extract the most informative features from different human body parts. Furthermore, we propose a MetaModel (MM) that effectively integrates the outputs of the ensemble architecture members. As a result, overall performance is enhanced, and comprehensive experiments are conducted on three benchmark datasets: CASIA-B [27], CASIA-C [28], and Outdoor-Gait [29]. The following steps are carried out in sequence: (1) The GEI is partitioned into five distinct horizontal parts. (2) To enhance the robustness of the features extracted from each part, n different homogeneous CNN networks are trained separately for each part. An MM is then trained for each of n different features, each of dimension d, extracted from the homogeneous networks. The most distinctive feature with dimension d is then produced for the current part through the MM. (3) The extracted five distinctive features are combined to achieve robust recognition. (4) A part-removal process is performed by calculating an absolute difference image over GEI with and without variations to handle different appearance-based variations. According to the calculated absolute difference image, a specific threshold value is determined for each horizontal part, and the horizontal parts above this threshold value are removed from the recognition process.

In the remaining part of the paper, Sect. 2 presents the related work. Section 3 describes the proposed method in detail. Experiments and results are presented in Sect. 4, and the last section concludes the paper.

2 Related work

Previous studies on gait recognition can be grouped into model-based and appearance-based approaches. The first group, model-based approaches, focuses on extracting the skeleton or structural features of the human body. They reconstruct a 3D model of the human body [11, 12] utilizing gait videos captured from multiple cameras at different view angles. For example, Liao et al. [13] proposed PoseGait, which leverages 3D human body joints as a feature for a CNN architecture. On the other hand, Teepe et al. [14] introduced GaitGraph, which combines skeleton poses with a Graph Convolutional Network (GCN). Recently, Li et al. [15] presented Cyclegait, which utilizes sequences of skeleton coordinates. Model-based methods are highly robust to cross-view variations; however, reconstructing a 3D model is complex and challenging in real-world scenarios due to the requirement for specialized cameras.

On the other hand, in appearance-based approaches, gait patterns are usually generated based on the gait period [1–4, 16, 43]. Among the generated gait patterns, the most commonly used gait pattern is GEI, which represents the dynamic and physical features in an image [1]. CGI, GEnI, GFI, and PEI are other gait patterns generated similarly to GEI. Motivated by the remarkable achievements of CNNs in recent years, CNN-based solutions have been increasingly applied in gait recognition [59, 60]. Wu et al. [8] applied 3D convolutions to extract spatio-temporal features from GEIs and learn pair-wise similarities, aiming to develop a general descriptor for human gait. Similarly, Wolf et al. [20] presented a 3D CNN model using a custom input format consisting of the grayscale image and optical flow in the horizontal and vertical directions. Takemura et al. [17] designed a Siamese network with a pair of input GEIs by optimizing the input, output, and loss functions of CNNs. Chao et al. [6] introduced an end-to-end deep learning model GaitSet, which extracts gait features from independent silhouettes. This architecture primarily consists of an operation called Set Pooling (SP), which aggregates frame-level features into a single set-level feature, and a structure called Horizontal Pyramid Mapping (HPM), which maps the set-level feature into a more discriminative space. In another study, Zhao et al. [51] used GaitSet as the backbone network and proposed a multi-modal network that uses pose heatmaps along with silhouettes to simultaneously gather part and body motion information. Elharrouss et al. [18] utilized GEIs for angle estimation and gait recognition using multitask CNN models. They first extracted GEIs by applying background subtraction techniques to detect moving objects. Then, the multitask CNN estimated the gait angle and performed the recognition process. Hou et al. [19] proposed a network named Gait Lateral Network (GLN), which can learn discriminative and compact representations from the silhouettes and leverage the inherent feature pyramid in deep CNNs to enhance the gait representations. However, the aforementioned studies focus on gait representations based on global information, which are more sensitive to appearance variations and may neglect fine-grained body part details. Therefore, numerous studies have utilized local features rather than global features to improve the robustness of gait feature representation. For instance, Fan et al. [10] introduced GaitPart, a model that captures part-level spatial features and local micro-motion features by employing a focal convolution layer. The model has two components a Frame-level Part Feature Extractor (FPFE) to learn the part-level spatial features and a Micromotion Capture Module (MCM) to model the local micro-motions features. Zhang et al. [7] proposed a temporal attention model to extract distinct features from four horizontal parts of silhouettes. In the model, multiple separate hybrid CNN- Long Short-Term Memory (LSTM) architecture is trained for each local body part. Lin et al. [5] presented a 3D CNNs framework to attain discriminative representations from both global and local information of the gait frames. They developed a Global and Local Feature Extractor (GLFE) module to aggregate global and local features in a principal manner and a new operation, namely Local Temporal Aggregation (LTA), to further preserve spatial information. Sepas-Moghaddam and Etemad [53] proposed a model that first extracts gait convolutional energy maps (GCEM) from frame-level convolutional features. The model then used an attentive recurrent model to learn the relations between different partial representations from split bins of the GCEM. In another study [54], a capsule network is adopted to learn deeper part-whole relationships. The network first obtains multi-scale partial representations and then recurrently learns the correlations and co-occurrences of the patterns among the partial features in forward and backward directions using Bidirectional Gated Recurrent Units (BGRU). Li et al. [52] introduced GaitSlice, which uses a Slice Extraction Device (SED) to form top-down inter-related slice features and a Residual Frame Attention Mechanism (RFAM) to acquire and highlight keyframes temporally. Thus, they analyzed human gait based on spatio-temporal slice features. Ma et al. [55] proposed a new framework that actual gait features include global motion patterns in multiple key regions and each global motion pattern is composed of a series of local motion patterns. They created a dynamic attention mechanism between the features of neighboring pixels that not only adaptively focuses on key regions but also generates more expressive local motion patterns. Chen et al. [56] proposed a framework called GaitAMR, which employs a holistic and partial temporal aggregation strategy to extract body movement descriptors both globally and locally. In the view domain, GaitAMR treats features selected from the optimal view as complementary information to the spatiotemporal domain. Meanwhile, in the temporal domain, it focuses on capturing rich and unique temporal associations in the subject's motion trajectory, both globally and partially. Wei et al. [57] introduce a Multi-scale Network for Gait Recognition (GMSN), which comprises two key modules: a Multi-scale Feature Extractor (MSFE) and a Part-based Horizontal Mapping (PHM). The MSFE utilizes parallel multi-scale CNNs to extract features at different scales, capturing local and global information for a more comprehensive gait representation. Meanwhile, the PHM enhances the learning of critical body parts, emphasizing clear contours and movement patterns. In another study, Pan et al. [58] designed Local Relation Convolutional (LRConv) layers, which extract local detailed gait features by preserving edge detail information and leveraging contextual correlation, along with the Human Body Focusing (HBF) module, which focuses on gait parts that are less affected by clothing occlusion. Xiong et al. [41] used a multimodal co-learning-based network to mine the complementary information of the silhouette and skeleton modalities. They abstracted parts of the human body and the motion linkages between parts or joints as graphs. It can be observed that the primary objective of these methods is to design an architecture that can most effectively achieve gait recognition, either by extracting global and local features more efficiently or by modeling the relationships among the extracted local features in the most meaningful way. They also generally do not directly focus on local areas affected by appearance-based variation. In this study, we introduce an ensemble learning method to extract the most effective and robust features from local regions in a novel way. First, we utilize horizontal GEI segments for recognition, then apply ensemble learning to the CNNs responsible for each local partition of the GEI to improve the overall network performance. In addition, we propose a part-elimination process for regions affected by appearance-based variations. Furthermore, we assess the effectiveness of this approach across three distinct datasets: CASIA-B [27], CASIA-C [28], and Outdoor-Gait [29].

3 Methodology

The Gait Energy Image (GEI) [1] is an effective and efficient gait pattern that captures both spatial and temporal characteristics of human movement. Compared to other gait representations, GEI reflects the overall shapes of gait silhouettes and their changes within a cycle while being relatively insensitive to contour distortions and small-scale changes [21]. In this study, the GEI pattern is divided into five horizontal parts to focus on local regions. Subsequently, ensemble learning is applied to extract features from each part, leading to more robust feature representations.

3.1 Ensemble learning

Ensemble learning aims to enhance prediction accuracy by combining multiple models. The basic idea is to obtain better results by using the strengths of different models. In this approach, different models or variations of the same model are employed to evaluate data from multiple perspectives. The predictions from individual models are aggregated using voting, averaging, or more advanced fusion techniques [23]. Traditional ensemble learning involves integrating conventional machine learning models and applying them across various domains. Due to its generalizability and ability to improve classification performance, ensemble learning is also employed in deep learning-based classification tasks. Usman et al. [38] proposed a deep learning-based ensemble learning method to predict epileptic seizures. They used a feature set extracted from Electroencephalogram (EEG) signals to train an ensemble classifier that combines the outputs of three different models using a model-agnostic meta-learning method. Kazmaier et al. [39] applied several heterogeneous ensemble learning techniques to classify sentiment analysis datasets in another study. In their work, a novel ensemble selection approach was introduced that avoids storing individual predictions and retraining all candidate models for an ensemble.

The general framework of any ensemble learning system is to use an aggregation function G to combine a set k of baseline classifiers, $c_1, c_2, \ldots, c_k$, towards predicting a single output. Given a dataset D of size n and features of dimension m, $D = \{(x_i, y_i), 1 \leq i \leq n, x_i \in R^m\}$, the prediction of the single output y_i by this ensemble learning is given as follows [23]:

$$y_i = \varphi\ (x_i) = G(c_1, c_2, \ldots, c_k) \tag{1}$$

The selection of a data sampling method is one of the most important factors affecting the performance of ensemble learning. In an ensemble learning architecture, ensuring diversity in the data sampling decisions of the base classifiers is crucial [23]. An example of an ensemble learning architecture consisting of three base classifiers trained on different data samples is presented in Fig. 1.

Figure 1 illustrates a general ensemble learning architecture with three base classifiers. In this architecture, three separate training datasets are sampled from the original dataset to ensure diversity. Each training dataset is used to train a separate classifier. The classifier outputs are then aggregated (with voting, averaging, etc.) to produce the ensemble modeling output.

Ensemble learning has gained increasing attention in deep learning; however, its application in gait recognition remains limited. Previous methods [21, 22] that employ ensemble learning for gait recognition exhibit certain methodological limitations and have not demonstrated consistent improvements on benchmark datasets. In this study, part-based gait recognition is performed using ensemble learning. To achieve diversity in data sampling, we employ two essential methods: k-fold cross-validation and bootstrap aggregating, or bagging. Additionally, detailed experiments are conducted on three different datasets: CASIA-B [27], CASIA-C [28], and Outdoor-Gait [29].

3.2 Part-Based gait

Part-based gait recognition has been investigated in many studies before, based on the observation that specific body parts exhibit unique movement characteristics, contribute differently to identification, and cause a decrease in recognition performance when covered. For this purpose, Rida et al. [24] presented a method to select the most discriminative horizontal body parts by leveraging Group Lasso on motion features, aiming to reduce intra-class variation and enhance recognition performance. On the other hand, Rokanujjaman et al. [25] evaluated five horizontal body parts and identified three as effective contributors to identification due to their positive effects while classifying the remaining two as redundant because they negatively impacted recognition. In another study [26], an adaptive outlier detection method was proposed, where each row of the gallery and probe samples was evaluated as valid or invalid based on the adaptive threshold determined for that row, and body parts affected by clothing variations were excluded. In this study, to exclude appearance-based variations from recognition, GEI is first divided into five horizontal parts, and then affected parts by clothing and carrying variations are detected and removed. Subsequently, the recognition process is conducted using only the remaining unaffected body parts. Figures 2 and 3 show two examples of part-removal processes

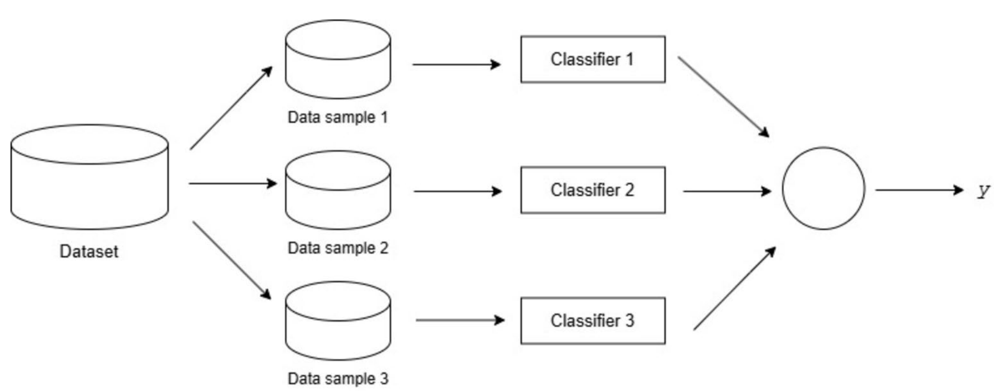

Fig. 1 An example of an ensemble learning architecture consisting of three base classifiers trained on different data samples [39]

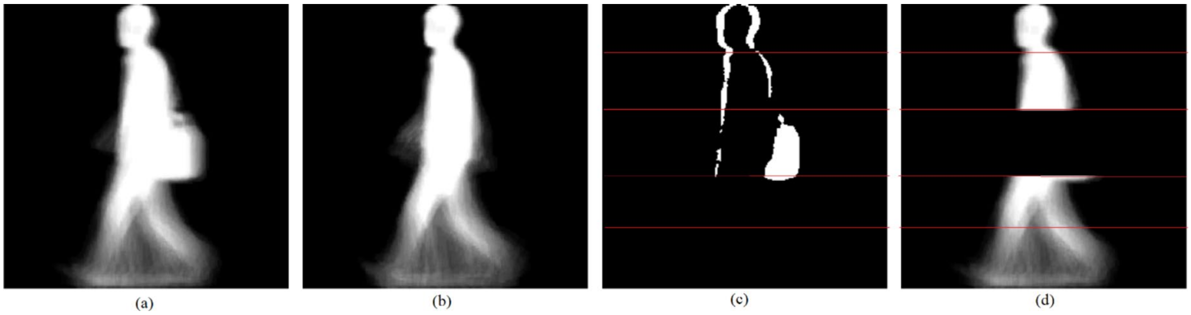

Fig. 2 Part-removal process for carrying variation from the CASIA-B **a** sample with carrying variation **b** sample with normal variation **c** absolute difference image **d** part-removed image

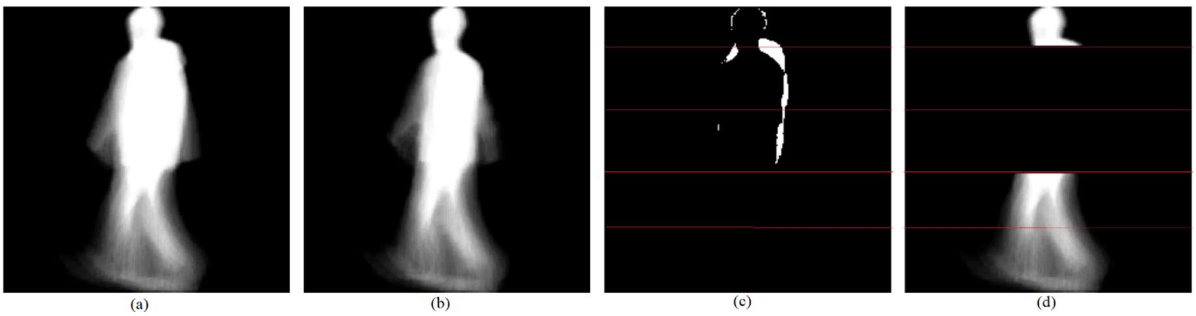

Fig. 3 Part-removal process for clothing variation from the CASIA-B **a** sample with clothing variation **b** sample with normal variation **c** absolute difference image **d** part-removed image

from the CASIA-B [27] dataset under carrying and clothing variation, respectively. In Fig. 2, a single part is detected and removed, whereas in Fig. 3, two parts are detected and removed.

In Figs. 2 and 3, an absolute difference image (marked with c) is initially calculated between a GEI with variations (marked with a) and its corresponding GEI without variations (marked with b). The resulting difference image is then divided into five horizontal parts, and an individual threshold value is determined manually depending on the type of variation for each part (For example, in the experimental study, th_1:1100 th_2:500 th_3:400 th_4:500 th_5:500 are the threshold (th) values of five parts for BG variation of the CASIA-B dataset). The threshold value is calculated based on the sum of pixel values in the local region. The parts exceeding their respective thresholds are considered the most affected by the variation. These parts are removed and replaced with fully black parts. Subsequently, recognition is conducted using only the remaining unaffected body parts. For example, in Fig. 2, recognition is performed on four body parts, and in Fig. 3, on three body parts.

3.3 Part-Based gait recognition with ensemble learning

Different parts of the human body demonstrate unique morphological characteristics and movement patterns during walking. Furthermore, partial features associated with specific body regions offer detailed information that is critical for individual identification tasks. Consequently, each body part requires a distinct representation [10]. Ensemble learning helps mitigate the limitations of individual models by leveraging diverse feature representations, leading to improved generalization. The body part representations can be further enhanced through ensemble learning to produce more robust and distinctive ones. As a result, the fusion of these representations enhances the discriminative power of the model, leading to more accurate recognition. In this paper, ensemble learning is applied to five distinct CNN architectures, each responsible for one of the five horizontally partitioned GEIs. This approach enhances the robustness of the extracted features and improves recognition accuracy through feature fusion. As illustrated in Fig. 4, the GEI is initially divided into five horizontal parts. Subsequently, each part is processed by a dedicated CNN model, which is subjected to ensemble learning independently. During each ensemble learning process, n data samples are generated, and n homogeneous CNN models are trained for each part. Each CNN output is represented by n distinct

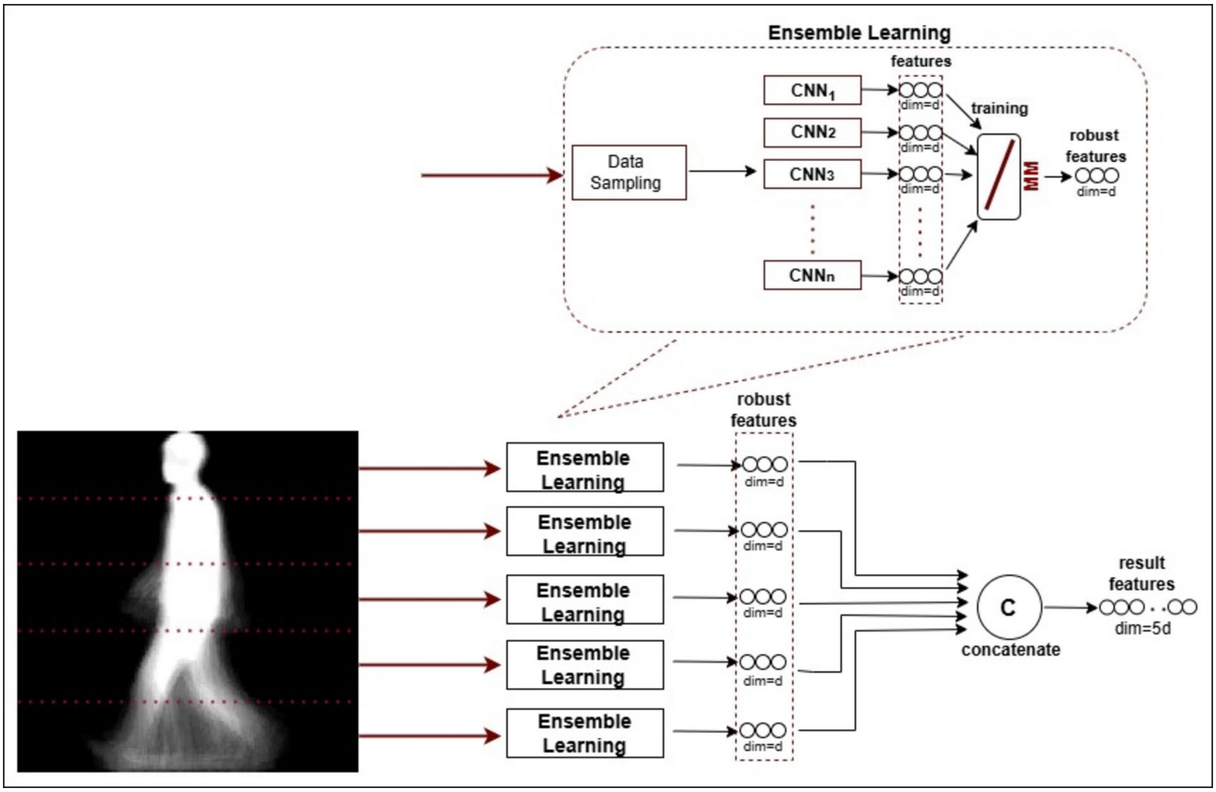

Fig. 4 The proposed gait recognition process is achieved through the concatenation of robust features extracted from five human body parts, which are further enhanced by ensemble learning

d-dimensional features. The diversity introduced by the data sampling process enhances the strength of these representations. The MM is used to fusion the n distinct d-dimensional features. Feature fusion aims to integrate multiple feature representations into a unified and more informative representation, enhancing the performance of recognition or classification tasks. The proposed MM employs a fully connected layer to combine feature vectors extracted from multiple independent models. By aggregating features from different sources, the model effectively captures complementary information, leading to a more discriminative representation. The linear transformation applied through the fully connected layer enables the model to learn optimal feature mapping, preserving essential characteristics while reducing redundancy. Unlike traditional ensemble methods that rely on majority voting or weighted averaging, this approach directly refines the fused feature space, allowing for more robust decision-making. Finally, recognition is achieved by concatenating the five robust and enriched body part representations. The framework of part-based gait recognition based on ensemble learning is described in detail in Algorithm 1.

Training:	Input: The GEI features
	Output: The rank-1 accuracy
	1: *Perform the part-removal process.* Parts with appearance-based variations are excluded from the training phase through a part-removal process.
	2: *Segment GEI into five horizontal parts*.
	3: *Data Sampling*: Create n data samples with k-fold cross-validation or bootstrap aggregating for each horizontal part.
	4: *Train ensemble learning architecture*. For each part, n homogeneous CNNs are trained.
Testing:	5: *Train MM*: Train the MM with a single linear layer on the outputs of n homogeneous CNNs.
	6: *Create features*: Create the most distinctive features produced from MM for all parts
	7: *Fuse*: Concatenate all part-based features.
	8: *Calculate accuracy*: Compute rank-1 accuracy for related views.

Algorithm 1 Part-based gait recognition based on ensemble learning

The first step of Algorithm 1 involves the part-removal process. By excluding parts affected by appearance-based variations from training, the classifiers can improve their generalization performance. The GEI is then segmented into five horizontal parts, each of which is processed independently. During data sampling, k-fold cross-validation or bootstrap aggregating is employed to ensure diversity for each horizontal part. In the k-fold cross-validation approach, the dataset is divided into k equally sized folds and k classifiers are trained iteratively on the remaining k-1 folds. In our scenario, k corresponds to the data sample number n. Bootstrap aggregating, on the other hand, applies random sampling with replacement to obtain n training data samples. This approach allows an instance to appear in multiple training data samples. The ensemble learning framework is then employed, where multiple CNNs are trained to produce n d-dimensional features for each horizontal part. These CNNs are identical in architecture but trained on different data samples. During the testing phase, the outputs of these CNNs represented as n distinct d-dimensional features, are utilized to train the MM, which consists of a single linear layer with an input size of $n \times d$ and an output size of d that transforms and refines the extracted features. The most distinctive features produced by the MM for each part are then combined to create a more discriminative representation. By concatenating these part-based features, a comprehensive gait representation is obtained, which is subsequently used to compute the rank-1 accuracy for different viewpoints. This approach effectively captures localized gait characteristics while leveraging ensemble learning to improve classification performance, ensuring robustness across various conditions. In the proposed architecture, the ConvNeXt [30] is employed as the backbone CNN architecture due to its demonstrated success in a gait recognition study [31]. The study showed that combining GEIs from two distinct body parts within a ConvNext-based multimodal network yielded superior performance compared to other multimodal networks based on different architectures. Furthermore, transfer learning is utilized to further enhance performance. For data sampling, the parameter n is set to 5, which corresponds to the total number of classifiers trained for a single horizontal part. Additionally, the dimensionality d is 1024, which represents the output size of the final layer of ConvNeXt.

In appearance-based variations, each part is evaluated for its presence. If a part is absent, no feature extraction is performed for that part. Instead, features are extracted solely from the available parts, and recognition is based exclusively on these extracted features. An example of this process is presented in Fig. 5.

Figure 5 illustrates the recognition process in the presence of missing human body parts. In the figure, the gallery image consists of a set of five horizontal parts (P1–P5), while the probe image contains three horizontal parts (P1, P4, P5), some of which are missing due to the part-removal process. Feature extraction is performed only for the available parts, omitting any missing components. First, the most discriminative part-based features are extracted from all available gallery and probe image parts using ensemble learning. The extracted features are then compared against the gallery, where a successful match is indicated with a checkmark (✓) and a failure with a cross (✖). Then, the features (F2, F3) corresponding to the missing parts in the probe image are discarded from the gallery image. Finally, recognition is performed by concatenating (©) the features of the three available horizontal parts, and the labeling process considers only successfully matched features, leading to a final recognition result.

4 Experiments and results

4.1 Datasets

In this section, we evaluate the proposed method's performance for efficient gait recognition on three distinct gait datasets: CASIA-B [27], CASIA-C [28], and Outdoor-Gait [29].

The *CASIA-B* [27] dataset contains 124 subjects, with 10 video groups for each. Six of these groups correspond to normal walking (NM), two to walking with a bag (BG), and

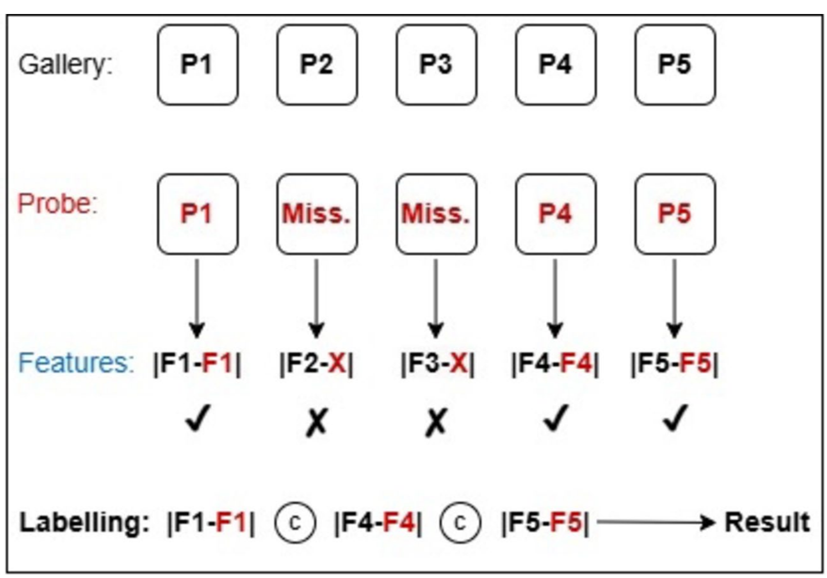

Fig. 5 An example of recognition process for appearance-based variations over existing horizontal segments

two to walking in a coat (CL). Each group contains 11 gait sequences captured from 11 different views (0º, 18º,., 162º, 180º) at fixed 18-degree intervals. For the experiments, 74 subjects are selected as the training set, and the remaining 50 subjects are used for the testing set. All gait data in the training set are utilized for training. In the test set, the NM#01-NM#04 sequences are considered as the gallery set, whereas the other sequences NM#05–06, BG#01–02, and CL#01–02 are assigned to the probe sets, respectively. The experimental settings for the CASIA-B dataset are shown in Table 1.

The *CASIA-C* [28] dataset consists of images gathered using a thermal (infrared) camera. There are 153 subjects walking under four distinct conditions: normal walking (fn), slow walking (fs), fast walking (fq), and walking with a bag (fb). Each subject has 10 gait sequences, comprising four sequences for normal walking (fn), two for slow walking (fs), two for fast walking (fq), and two for walking with a bag (fb). During the training phase, the first 24 and 62 subjects are used as the training set, while the remaining 53 subjects are used as the testing set. For each subject, the nm sequences are designated as the gallery set, whereas the fs, fq, and fb sequences are used as the probe sets, respectively.

Outdoor-Gait [29] is a detailed dataset characterized by its complex outdoor scenes. The dataset consists of 138 individuals under three clothing conditions (NM: normal, CL: wearing a coat, BG: carrying a bag) across three distinct scenes (SCENE-1: simple background, SCENE-2: static but complex background, SCENE-3: dynamic and complex background with moving objects). For the experiments, we use the first 69 individuals for the training set and the remaining individuals for the testing set. Each condition includes at least two video sequences in both the gallery and probe sets.

4.2 Implementation details

During the implementation phase, the GEI feature is utilized for three distinct datasets: CASIA-B, CASIA-C, and Outdoor-Gait. In the GEI construction stage, a silhouette preprocessing procedure is initially performed. To mitigate the effects of variations in distance and view angle, two primary operations are applied: scale normalization and central alignment [42]. Subsequently, GEI features are generated by averaging properly aligned and preprocessed silhouettes. Finally, each GEI is segmented into five horizontal parts, and then each part is normalized to a size of 224×224.

Table 1 The experimental settings for the CASIA-B dataset

Training set	Testing set	
	Gallery	Probe
ID: 001–074	ID: 075–124	ID: 075–124
Seqs: NM #01–06, BG #01–02, CL #01–02	Seqs: NM #01–04	Seqs: NM #05–06, BG #01–02, CL #01–02

Figure 6 presents examples of GEIs obtained from CASIA-B, CASIA-C, and Outdoor-Gait datasets. Figure 6a illustrates GEIs from the CASIA-B dataset under NM, BG, and CL variations across 11 different views. Similarly, Fig. 6b displays GEIs generated from the CASIA-C dataset under fn, fs, fq, and fb conditions for both genders. Finally, Fig. 6c shows GEIs from the Outdoor-Gait dataset under NM, BG, and CL variations across three distinct scenes (SCENE-1, SCENE-2, and SCENE-3) for both genders.

In the training stage, the batch size is fixed at 4, and the learning rate is set to 1e-4 for all datasets. In the CASIA-B dataset, the number of epochs is set to approximately 30 on average for each body part. For the CASIA-C dataset, the number of epochs is set to approximately 45 for 24 subjects and around 50 on average for 62 subjects. Similarly, in the Outdoor-Gait dataset, the number of epochs is set to approximately 50 on average for each body part. When determining the number of epochs for ensemble learning, fewer epochs are assigned for *k*-fold cross-validation, whereas a higher number of epochs is assigned for the bootstrap aggregating approach. All experiments use Stochastic Gradient Descent (SGD) with momentum = 0.9 as the optimizer and utilize cross-entropy loss. During the test phase, a single d-dimensional feature is derived by training the MM on n d-dimensional features extracted from each part, where d is 1024, and n is set to 5. Cosine distance is selected as the similarity metric.

4.3 Experiments on CASIA-B dataset

The proposed architecture is first evaluated on the CASIA-B dataset. In Table 2, a comprehensive comparison is provided by considering the GEI with two different configurations, partitioning the body horizontally into four and five parts. Additionally, five-part learning and five-part learning with ensemble methods are evaluated separately. Finally, the comparison results incorporating the part-removal processes, implemented to mitigate the effects of BG and CL variations during the training and testing stages, are also presented in separate rows.

Although some previous studies [7] have suggested that 4-part learning (4part) represents the most optimal segmentation strategy, Table 2 demonstrates that 5-part learning (5part) achieves better recognition rates within the proposed architecture. This difference, while less noticeable in the NM variation, becomes more evident in the BG and CL

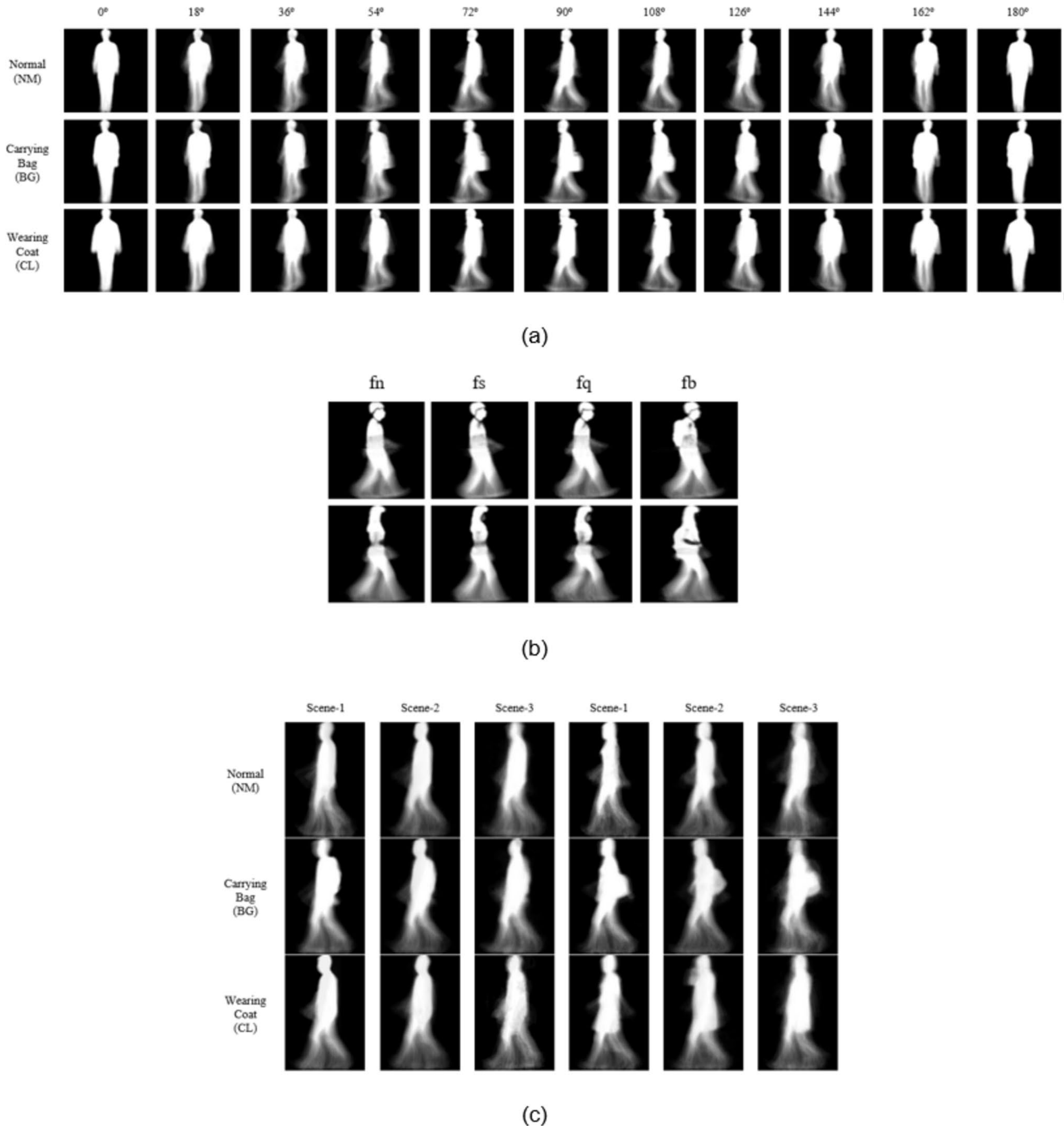

Fig. 6 Some examples of GEI. GEIs from the **a** CASIA-B [27] **b** CASIA-C [28] **c** Outdoor-Gait [29] datasets

variations. The increase in the recognition rate has exceeded 5%, particularly in the CL variation. This improvement can be attributed to the compression of the region impacted by the variation within a narrower area and an increase in the number of unaffected parts. Furthermore, the recognition rate is enhanced significantly when the part-removal process (5part+rmv) is applied during the training and testing phases of 5part. This improvement is particularly noticeable under the CL variation, where the increase exceeds 10%. This enhancement can be due to the elimination of deformed body segments caused by variation from the training process, leading to more stable learning, and the more consistent similarity calculations during the testing phase. Additionally, the application of the part-removal process helps to eliminate body segments containing unrelated

Table 2 Comparison of different part-based learning architectures on the CASIA-B dataset in terms of rank-1 accuracy (%) (excluding identical views)

Gallery NM#1–4		0º -180º											avg
Probe		0º	18º	36º	54º	72º	90º	108º	126º	144º	162º	180º	
NM #5–6	4part	83.5	93.8	95.6	95.7	90.5	85.6	90.5	92.2	95.3	91.4	81.5	90.5
	5part	84.7	93.9	96.9	94.7	89.6	86.2	88.9	92.8	95.6	91.8	83.2	90.8
	5part + rmv	85.3	92.7	96.9	**96.1**	90.8	87.1	89.9	**94.6**	95.8	93.3	83.1	91.4
	5part + Ens(Kfold)	87.7	**95.5**	**97.9**	95.7	**92.5**	88.3	89.8	93.2	**96.9**	94.2	84.2	92.3
	5part + Ens(Boots)	86.5	94.9	96.9	94.1	89.8	87.3	89.7	92.5	95.3	93.6	84.3	91.3
	5part + rmv + Ens(Kfold)	87.3	94.8	97.5	96.0	92.0	**88.6**	90.6	93.7	**96.9**	94.0	85.1	**92.4**
	5part + rmv + Ens(Boots)	**88.3**	94.6	97.3	94.9	90.8	88.5	**90.7**	93.4	96.1	**94.6**	**85.7**	92.3
BG #1–2	4part	79.2	86.6	87.7	85.4	78.8	73.1	77.2	83.9	86.9	81.8	75.0	81.4
	5part	80.0	88.5	86.2	83.2	77.4	74.2	79.5	86.0	89.8	85.6	77.1	82.5
	5part + rmv	78.1	88.1	88.4	87.2	80.0	77.0	79.0	86.6	90.5	84.8	75.8	83.2
	5part + Ens(Kfold)	**80.1**	88.4	88.8	87.0	78.8	73.6	78.1	86.6	90.7	85.9	78.2	83.3
	5part + Ens(Boots)	78.8	88.6	**89.5**	86.4	79.7	73.7	79.5	85.8	89.4	**86.7**	77.8	83.3
	5part + rmv + Ens(Kfold)	79.6	87.5	87.9	**87.6**	80.0	76.5	81.1	**86.7**	91.2	86.0	**78.9**	83.9
	5part + rmv + Ens(Boots)	77.9	**89.4**	89.3	87.1	**80.1**	**78.1**	**82.2**	86.6	**92.0**	85.9	78.6	**84.3**
CL #1–2	4part	37.7	48.5	51.9	53.8	51.1	46.4	46.4	42.8	42.1	39.9	32.2	44.8
	5part	46.1	56.3	61.6	60.4	55.5	50.4	53.1	47.5	49.4	47.9	40.0	51.7
	5part + rmv	52.5	68.1	**71.7**	67.0	**66.0**	**62.4**	**61.4**	**61.6**	**64.8**	**61.5**	46.5	**62.1**
	5part + Ens(Kfold)	43.0	54.0	56.9	55.8	50.4	46.4	45.0	43.4	44.7	45.1	38.5	47.6
	5part + Ens(Boots)	47.9	58.7	62.5	61.2	53.9	50.2	51.2	48.9	52.5	52.7	44.7	53.1
	5part + rmv + Ens(Kfold)	50.4	68.3	70.5	**67.4**	62.2	57.7	58.9	60.6	63.3	58.6	46.9	60.4
	5part + rmv + Ens(Boots)	**53.4**	**68.7**	68.0	65.0	61.9	58.4	60.1	61.0	63.2	59.2	**47.9**	60.6

The bold denotes the best value

objects, further enhancing the robustness and reliability of the model.

It can be understood from Table 2 that significant improvements are achieved by integrating ensemble learning into 5-part learning (5part+Ens). The recognition rates achieved with *k*-fold cross-validation (5part+Ens(Kfold)) and bootstrap aggregating (5part+Ens(Boots)) are markedly higher compared to standard 5-part learning (5part). Although 5part+Ens(Kfold) demonstrates superior recognition performance compared to 5part in the NM variation, this performance is equal to 5part+Ens(Boots) in the BG variation and lower in the CL variation. This decrease in the CL variation may be attributed to the fact that the number of samples unaffected by variation can be higher in the data samples obtained through random sampling with replacement, whereas it may be lower in the data samples generated through folds.

Table 2 reveals that incorporating the part-removal process into ensemble learning approaches (5part+rmv+Ens) further improves the recognition performance of 5part+Ens. This improvement has occurred at different rates for each variation in the *k*-fold cross-validation and bootstrap aggregating approaches. Such an effect is particularly more evident in the NM and BG variations of the bootstrap aggregating approach (5part+rmv+Ens(Boots)) and CL variation of the *k*-fold cross-validation approach (5part+rmv+Ens(Kfold)), with these improvements being 1%, 1%, and 12.8%, respectively.

In Table 2, when the effect of ensemble learning approaches is examined on 5-part learning with part-removal process (5part+rmv), the expected improvement is observed in the NM and BG variations, while a decrease of nearly 2% is observed in the CL variation. This is likely due to the higher number of affected parts by variation in CL and the limited contribution of ensemble learning to the remaining parts. In addition, it can be said that both ensemble learning approaches achieve recognition rates close to each other in all variations: NM, BG, and CL for 5part+rmv+Ens. Only the *k*-fold cross-validation approach slightly outperforms bootstrap aggregating for the NM variation. As a result, the 5part+rmv+Ens(Kfold) and 5part+rmv+Ens(Boots) methods achieve the best performance in the NM and BG variations, while the 5part+rmv method yields the highest accuracy in the CL variation. Among the ensemble approaches, the 5part+rmv+Ens(Boots) method achieves the best results only at 0º and 180º views in the CL variation.

Figure 7, illustrated from the results in Table 2, indicates that while all methods exhibit similar performance in the NM variation, the performance differences become more pronounced in the BG variation and present a clear divergence in the CL variation. This distinction arises from the impact of the part-removal process on variations, which affects the consistency of ensemble learning decisions and contributes to the observed performance differences.

Table 3 present a comparative analysis of the proposed method, both with and without ensemble learning incorporating part-removal, against other appearance-based and model-based methods in the literature. The results and Fig. 8 indicate that the proposed ensemble learning method incorporating the part-removal (5part+rmv+Ens) process outperforms the appearance-based CNN-LB, JCNN, and GaitNet methods that focus on global features, except the GaitSet method. Although GaitSet takes a set of gait silhouettes as input, it employs Horizontal Pyramid Pooling (HPP) to obtain the final representation, aiming to extract multi-scale part-level features. Thus, it gathers both local and global information on gait. On the other hand, the performance of the proposed method is lower than that of GaitPart, GaitGL, and GaitBase, which focus on local features. These methods are designed to effectively extract the most relevant local features specific to gait. For example, GaitPart employs the Focal Convolution Layer (FConv), which horizontally divides the input feature map into multiple parts to enhance the fine-grained learning of part-level spatial features. GaitGL introduces a novel GLFE module to extract more comprehensive features that incorporate both global and local information while capturing the relationships among neighboring parts. Finally, GaitBase employs a framework that horizontally divides the 3D feature map of silhouettes into multiple parts and projects them into a feature space that enhances their separability. The presence of modules designed specifically for gait patterns in these approaches has significantly improved the recognition rate. Nevertheless, the proposed method is less complex and more straightforward to implement. It provides a general framework that demonstrates the fundamental effect of ensemble learning on part-based methods without relying on gait-specific modules. The results indicate that the performance can be enhanced through the application of ensemble learning. Furthermore, enriching ensemble learning approaches with modules specifically designed for gait patterns could further improve performance.

Based on Table 3; Fig. 9, the proposed method demonstrates the best performance in the NM and BG variations while yielding comparable results to other methods in the CL variation when compared to model-based methods. Notably, compared to the previous state-of-the-art method, GaitGraph, the proposed ensemble learning approach achieves improvements of 4.6% in the NM variation and 9.5% in the BG variation, while attaining an accuracy of 60.6% in the CL variation. Similarly, compared to the state-of-the-art GaitDLF and MSRG methods, it demonstrates strong performance on the CASIA-B dataset under the NM and BG variations for the training set of 74 subjects. The

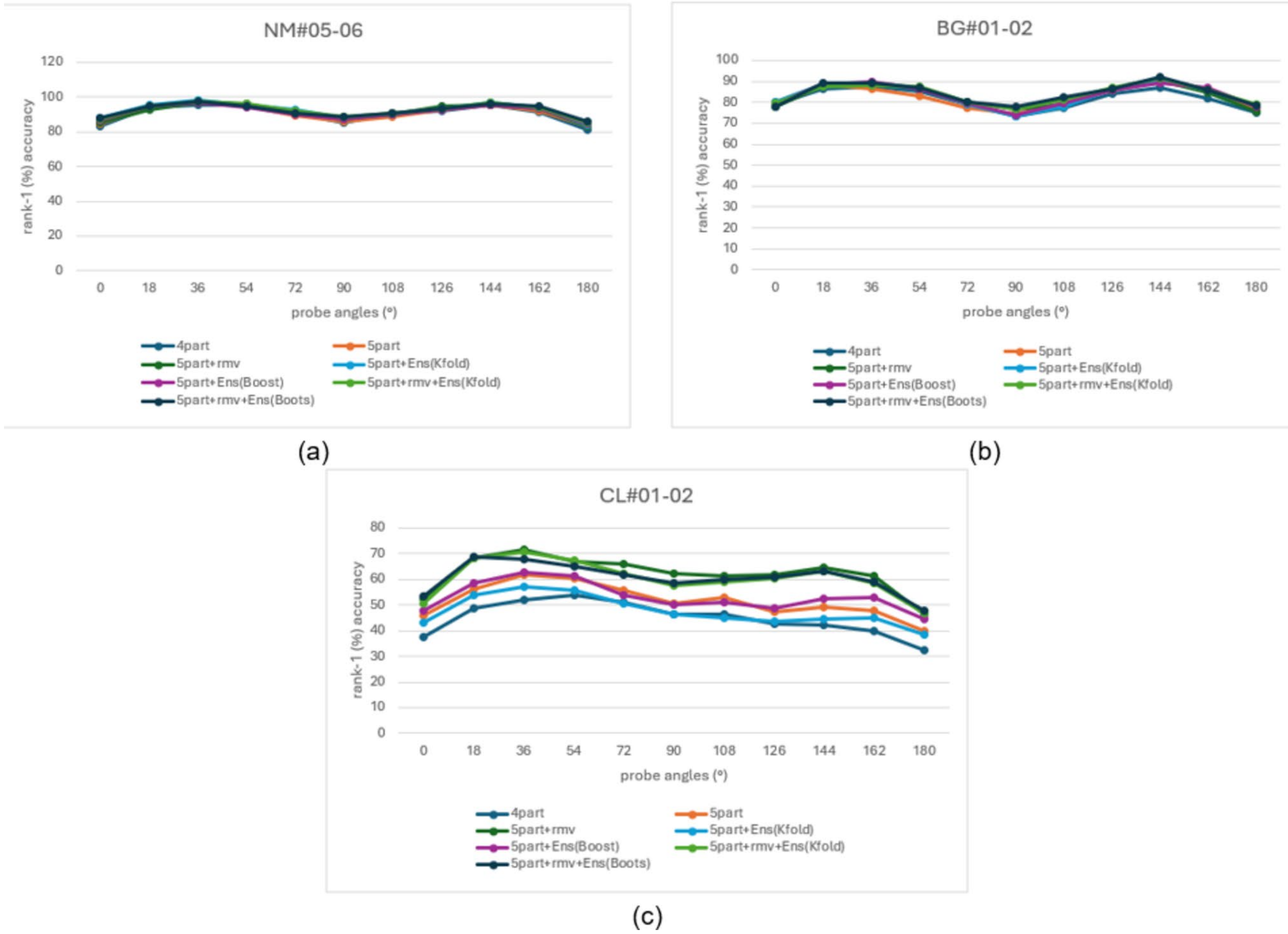

Fig. 7 Comparison of proposed part-based learning architectures on the CASIA-B dataset under appearance-based variations **a** NM **b** BG **c** CL

Table 3 Comparison of the proposed methods in terms of Avaraged rank-1 accuracy (%) with appearance-based and model-based methods on the CASIA-B dataset (excluding identical views)

Type	Methods	NM	BG	CL
Appearance-based	CNN-LB [8]	89.9	72.4	54.0
	JCNN [32]	91.2	75.0	54.0
	GaitSet [6]	95.0	87.2	70.4
	GaitNet [33]	91.6	85.7	58.9
	GaitPart [10]	96.2	91.5	78.7
	GaitGL [5]	97.4	**94.5**	**83.6**
	GaitBase [61]	**97.6**	94.0	77.4
Proposed	5part+rmv	91.4	83.2	62.1
	5part+rmv+Ens	92.3	84.3	60.6
Model-based	PoseGait [13]	68.7	44.5	36.0
	GaitGraph [14]	87.7	74.8	66.3
	GaitGraph2 [34]	82.0	73.2	63.6
	ResGait [35]	89.6	79.2	**70.2**
	GaitDLF [62]	84.9	70.7	67.9
	MSRG [63]	89.0	75.6	70.0
Proposed	5part+rmv	91.4	83.2	62.1
	5part+rmv+Ens	**92.3**	**84.3**	60.6

The bold denotes the best value

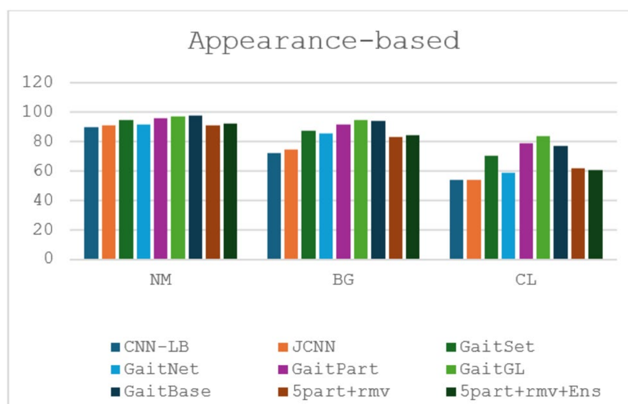

Fig. 8 Accuracy comparison of proposed methods with appearance-based methods on the CASIA-B dataset

limited improvement in the CL variation may be because skeleton features are more resistant to faults caused by the coat.

Table 4; Fig. 10 provide a comparative analysis of the proposed method, with and without ensemble learning, which incorporates the part-removal process alongside

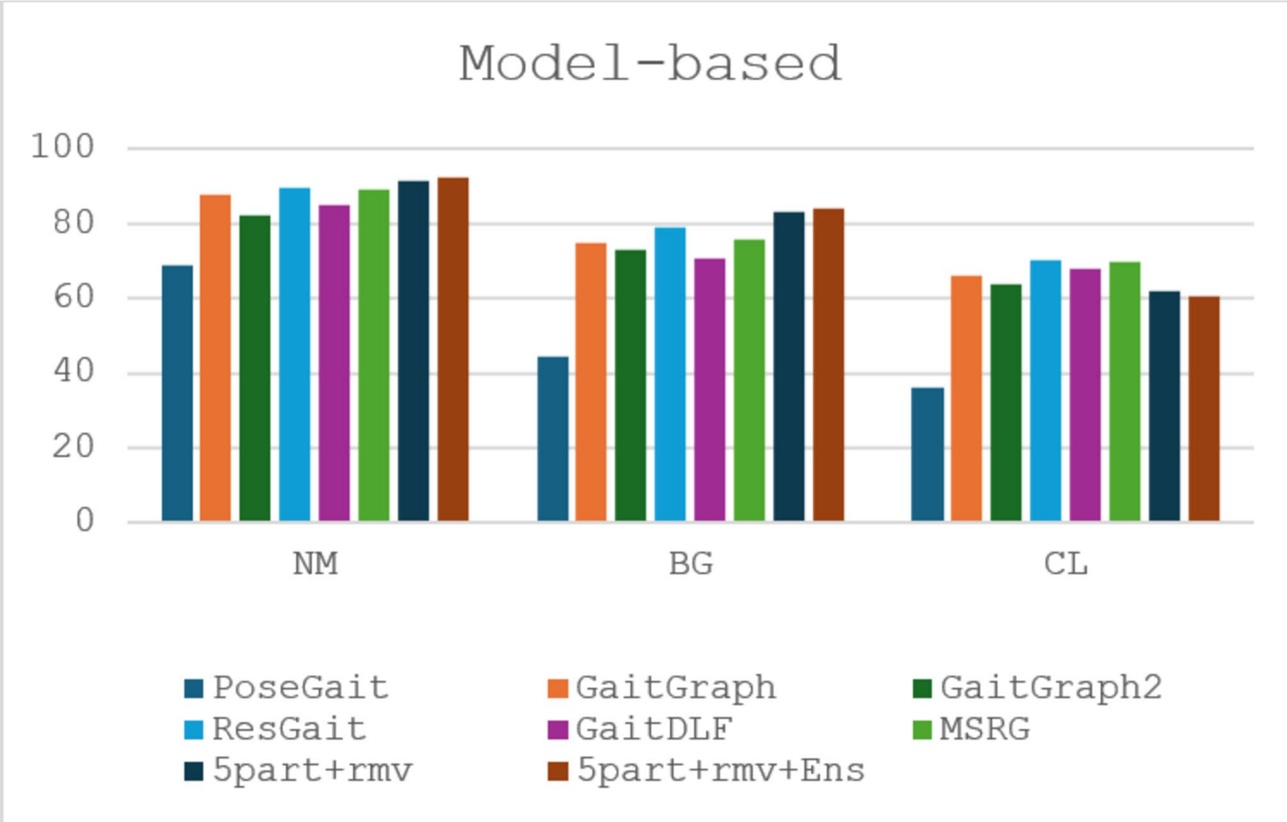

Fig. 9 Accuracy comparison of proposed methods with model-based methods on the CASIA-B dataset

previous methods under identical-view conditions. According to the results, all approaches exhibit very similar performance under the NM variation, regardless of whether they are trained with 62 or 74 subjects. The slight difference between the approaches may be attributed to the difference in the size of the training dataset. However, this difference becomes significantly more noticeable under the BG and CL variations. Although this may be partially attributable to the discrepancy in the number of subjects, the primary factor appears to be the differences in the methodological approaches employed. While PoseGait, GOFN, and UGaitNet employ a global approach, PSN offers a partially local approach by utilizing the local patterns of the joints. However, the 5part+rmv has significantly improved recognition performance in the BG and CL variations by leveraging the part-removal process and addressing the pattern of each horizontal part individually. When evaluating the impact of ensemble learning on the recognition performance of the 5part+rmv, it can be observed that similar to the results in Table 2, the part-removal process particularly affects the consistency of the ensemble learning decisions in the CL variation.

4.4 Experiments on CASIA-C dataset

The proposed ensemble learning architecture is secondly evaluated on the CASIA-C dataset. Table 5 presents the evaluation results of the proposed method and its comparison with other studies using different training sets. The *k*-fold cross-validation is chosen as the ensemble learning approach.

As presented in Table 5, the 5part+Ens achieved a rank-1 accuracy of 96.3% when trained on 24 subjects and 98.8% on 62 subjects, demonstrating the best performance. Additionally, an analysis of the table shows that ensemble learning significantly improves the recognition rate of 5part for both 24 and 62 subjects. Specifically, the average (avg) rank-1 accuracy value increase is 4.3% for the 24 subjects and 1% for the 62 subjects. Compared to other studies, the proposed ensemble learning method outperforms the PSN, GOFN, and GaitSTAR methods. This can be attributed to the fact that the PSN method follows a model-based approach, which may not be well-suited for thermal (infrared) images. Similarly, the recognition process in the GOFN and GaitSTAR methods, which

Table 4 Comparison of the proposed methods with previous methods under identical-view condition in terms of rank-1 accuracy (%) on the CASIA-B dataset

Gallery NM#1–4			0° -180°											avg
Probe			0°	18°	36°	54°	72°	90°	108°	126°	144°	162°	180°	
NM #5–6	62id	PoseGait [13]	96.0	96.8	96.0	96.8	96.0	97.6	97.6	94.4	96.8	97.6	97.6	96.6
		PSN [36]	99.2	99.2	98.4	99.2	97.6	98.4	97.6	96.8	97.6	97.6	99.2	98.1
		GOFN [40]	98.4	99.2	99.2	97.6	97.6	96.8	96.8	99.2	99.2	98.4	97.6	98.2
	74id	UGaitNet [9]	99	100	100	99	100	100	100	98	98	96	98	98.9
		5part + rmv	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**98**	97	98	98	**99.2**
		5part + rmv + Ens	**100**	**100**	**100**	**100**	**100**	99	**100**	**98**	**98**	**98**	**98**	**99.2**
BG #1–2	62id	PoseGait [13]	74.2	75.8	77.4	76.6	69.4	70.2	71.0	69.4	74.2	65.3	60.5	71.3
		PSN [36]	86.3	84.7	83.1	88.7	90.3	86.3	90.3	83.9	84.7	76.6	80.7	85.0
		GOFN [40]	84.7	88.7	92.7	91.1	85.5	80.7	87.1	88.7	91.9	91.1	79.9	87.5
	74id	UGaitNet [9]	98	97	95	94	88	90	88	91	93	95	91	92.7
		5part + rmv	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**98**	**98**	**98**	**98**	**99.3**
		5part + rmv + Ens	**100**	99	99	**100**	**100**	**100**	**100**	**98**	**98**	**98**	97	99.0
CL #1–2	62id	PoseGait [13]	46.8	48.4	57.3	61.3	58.1	56.5	59.7	54.8	55.7	58.1	39.5	54.2
		PSN [36]	64.5	68.6	70.2	71.0	68.6	64.5	62.9	56.5	59.7	59.7	60.5	64.2
		GOFN [40]	67.7	72.8	77.6	73.4	67.0	67.7	66.1	68.5	69.3	68.5	64.5	69.4
	74id	UGaitNet [9]	59	64	70	67	68	70	71	69	71	68	57	66.7
		5part + rmv	**97**	**97**	**98**	**95**	**98**	95	93	**93**	**94**	**95**	**94**	**95.4**
		5part + rmv + Ens	**97**	96	94	93	97	**96**	**94**	**93**	92	93	90	94.1

The bold denotes the best value

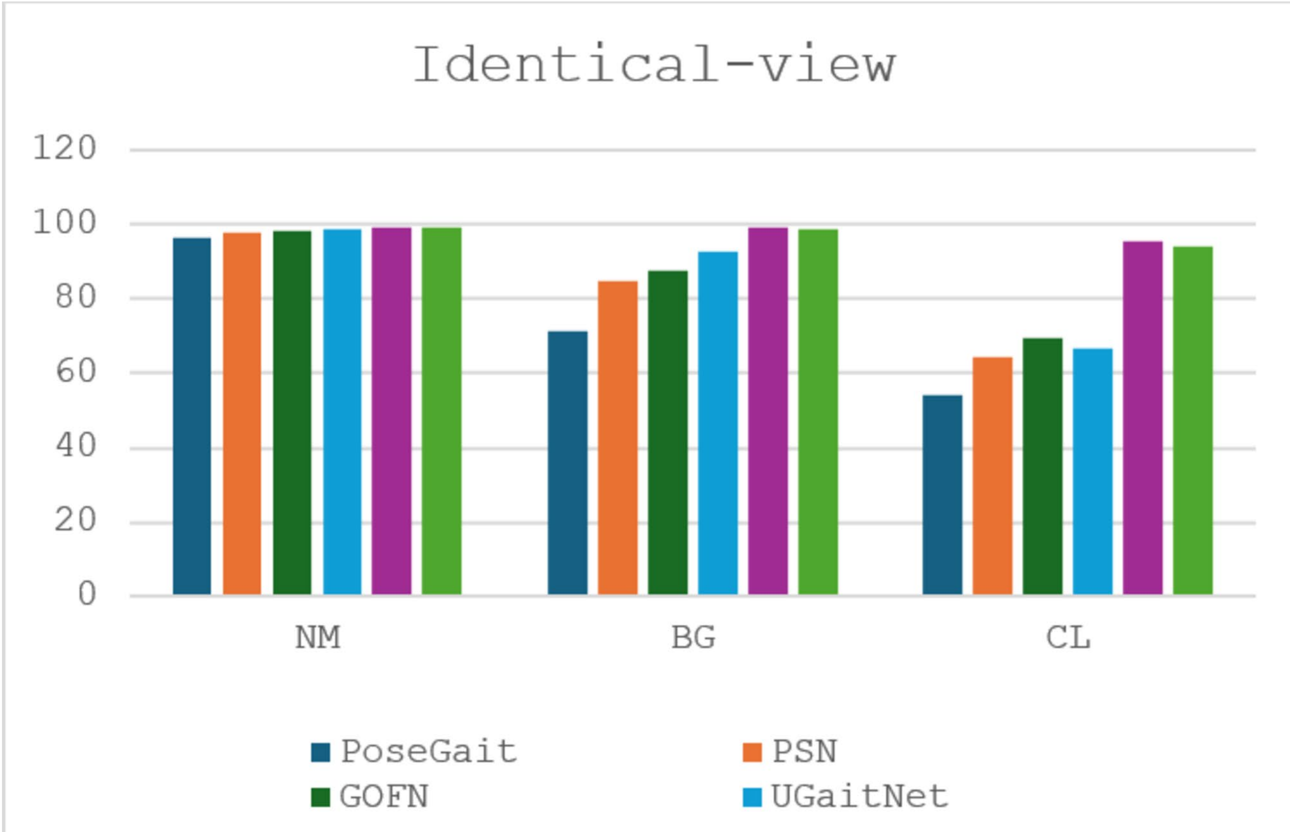

Fig. 10 Accuracy comparison of proposed methods under identical-view conditions

Table 5 Comparison of the proposed method in terms of rank-1 accuracy (%) with other methods on the CASIA-C dataset

Training	Methods	fb	fq	fs	avg
24id	PSN [36]	38.1	58.6	56.0	50.9
	GOFN [40]	39.5	64.2	54.3	52.7
	GaitSTAR [64]	40.7	65.3	56.4	54.1
	5part	84.2	96.3	95.4	92.0
	5part+Ens	**93.5**	**99.1**	**96.3**	**96.3**
62id	PSN [36]	42.3	63.6	60.0	55.3
	GOFN [40]	43.5	69.3	58.2	57.0
	GaitSTAR [64]	44.6	71.5	59.1	58.4
	5part	97.2	**99.1**	97.2	97.8
	5part+Ens	**99.1**	**99.1**	**98.1**	**98.8**

The bold denotes the best value

extract optical flow features from thermal images, may face similar limitations. In contrast, the proposed method benefits from GEI features, likely providing a more suitable input representation for thermal images. Additionally, parallel to the results in Table 4, the proposed method's ability to analyze each horizontal segment of the gait pattern separately has enabled it to achieve superior performance under identical-view conditions.

4.5 Experiments on uutdoor-gait dataset

The final evaluation of the proposed architecture is conducted on the Outdoor-Gait dataset. Table 6 shows a comparative analysis of the proposed method—evaluated both with and without ensemble learning—against the appearance-based

Table 6 Comparison of the proposed method with other methods on the Outdoor-Gait dataset

Gallery	NM			BG			CL			avg
Probe	NM	BG	CL	NM	BG	CL	NM	BG	CL	
GaitNet[29]	96.9	89.1	60.2	**92.0**	97.1	59.7	58.7	55.4	97.3	78.5
3Dhuman[37]	90.0	83.6	80.9	77.5	90.3	72.6	72.2	66.4	88.8	80.2
5part	**98.3**	90.8	**85.6**	88.4	98.1	74.6	**80.4**	72.9	97.8	87.4
5part+Ens	97.8	**91.1**	84.5	89.7	**98.5**	**77.4**	79.3	**75.1**	**98.3**	**88.0**

The bold denotes the best value

GaitNet [29] and the model-based 3Dhuman [37] studies on the Outdoor-Gait dataset. The *k*-fold cross-validation is chosen as the ensemble learning approach for 5part+Ens.

From Table 6, it is evident that the 5part+Ens approach achieves the highest recognition rate compared to other studies, GaitNet and 3Dhuman. Furthermore, ensemble learning contributes to an overall improvement in the average recognition rank-1 value (avg) for the 5part approach. However, in certain variations, ensemble learning does not enhance performance. For example, ensemble learning cannot improve the recognition rate when the gallery set is NM, the probe set is either NM or CL or when the gallery set is CL and the probe set is NM. Moreover, unlike the identical-view performance of CASIA-B reported in Table 4 and CASIA-C in Table 5—which, along with Outdoor-Gait, is an identical-view dataset—satisfactory performance could not be achieved across all gallery and probe conditions. This limitation might be attributed to the Outdoor-Gait dataset, which includes not only appearance-based variations but also complex background scene variations. This situation may have led the base classifiers to make significantly different decisions in ensemble learning, thereby preventing the extraction of robust features.

5 Conclusion

In this study, the performance of part-based gait recognition was improved through the application of ensemble learning. Specifically, the GEI features were partitioned into five horizontal parts, and a CNN model was trained for each local part. To further enhance the features generated by these CNNs, ensemble learning was employed, followed by the training of the MetaModel (MM), which aggregated the outputs of the ensemble CNNs to produce a single, more robust feature representation. Unlike traditional ensemble methods that rely on majority voting or weighted averaging, this approach refined the fused feature space directly, allowing for more robust decision-making. Furthermore, the MM model proved particularly advantageous in scenarios where multiple models provided diverse yet relevant information, enabling the extraction of a more comprehensive feature representation and contributing to the observed performance gains. On the other hand, a part-removal process was utilized to mitigate the effect of appearance-based variations. To select the part to be removed, a difference image was calculated between the images with and without variations. The part most affected by the variation was excluded from the training and testing phases using a predefined threshold value. Finally, the recognition process was conducted using the final set of part-based robust features. Experimental results on the CASIA-B, CASIA-C, and Outdoor-Gait datasets demonstrate that ensemble learning, combined with the part-removal process, significantly enhanced gait recognition performance. Compared to previous approaches, the part-based recognition with ensemble learning achieved superior performance, particularly on the CASIA-C and Outdoor-Gait datasets consisting of identical-view cases, while also yielding satisfactory results on the CASIA-B dataset. Notably, incorporating modules capable of modeling both global information and the relationships between local information, as present in the literature, would be effective in improving the proposed method for the CASIA-B dataset. This would also enhance the generalizability of the proposed method for gait recognition.

Author contributions Conceptualization, Methodology, Formal analysis and investigation, and Writing - original draft preparation stages were carried out by Büşranur Yaprak, while Eyüp Gedikli provided consultancy. All authors reviewed the manuscript.

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Data availability The datasets and source code are accessible at https://github.com/busrakckugurlu/Enhancing-Part-based-Gait-Recognition-via-Ensemble-Learning-and-Feature-Fusion/tree/main.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

References

1. Han J, Bhanu B (2005) Individual recognition using gait energy image. IEEE Trans Pattern Anal Mach Intell 28(2):316–322. https://doi.org/10.1109/TPAMI.2006.38
2. Wang C, Zhang J, Wang L, Pu J, Yuan X (2011) Human identification using Temporal information preserving gait template. IEEE Trans Pattern Anal Mach Intell 34(11):2164–2176. https://doi.org/10.1109/TPAMI.2011.260
3. Bashir K, Xiang T, Gong S (2009) Gait recognition using gait entropy image. In: Proceedings of the International Conference on Crime Detection and Prevention (pp.1–6)

4. He Y, Zhang J, Shan H, Wang L (2019) Multi-task GANs for viewspecific feature learning in gait recognition. IEEE Trans Inf Forensics Secur 14(1):1102–1113. https://doi.org/10.1109/TIFS.2018.2844819
5. Lin B, Zhang S, Yu X (2021) Gait recognition via effective global-local feature representation and local temporal aggregation. In: Proceedings of the IEEE/CVF international conference on computer vision (pp. 14648–14656)
6. Chao H, He Y, Zhang J, Feng J (2019) GaitSet: regarding gait as a set for cross-view gait recognition. In: AAAI Conference on Artificial Intelligence (pp. 8126–8133)
7. Zhang Y, Huang Y, Yu S, Wang L (2019) Cross-view gait recognition by discriminative feature learning. IEEE Trans Image Process 29:1001–1015. https://doi.org/10.1109/TIP.2019.2926208
8. Wu Z, Huang Y, Wang L, Wang X, Tan T (2017) A comprehensive study on cross-view gait based human identification with deep Cnns. IEEE Trans Pattern Anal Mach Intell 39(2):209–226. https://doi.org/10.1109/TPAMI.2016.2545669
9. Marín-Jiménez MJ, Castro FM, Delgado-Escaño R, Kalogeiton V, Guil N (2021) UGaitNet: multimodal gait recognition with missing input modalities. IEEE Trans Inf Forensics Secur 16:5452–5462. https://doi.org/10.1109/TIFS.2021.3132579
10. Fan C, Peng Y, Cao C, Liu X, Hou S, Chi J, Huang Y, Li Q, He Z (2020) Gaitpart: Temporal part-based model for gait recognition. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 14225–14233)
11. Nixon MS, Carter JN, Nash JM, Huang PS, Cunado D, Stevenage SV (1999) Auto-matic gait recognition. In: Motion Analysis and Tracking (p. 3/1–3/6)
12. Feng Y, Li Y, Luo J (2016) Learning effective gait features using LSTM. In: the 23rd International Conference on Pattern Recognition (pp. 325–330)
13. Liao R, Yu S, An W, Huang Y (2020) A model-based gait recognition method with body pose and human prior knowledge. Pattern Recognit 98:107069. https://doi.org/10.1016/j.patcog.2019.107069
14. Teepe T, Khan A, Gilg J, Herzog F, Hörmann S, Rigoll G (2021) Gaitgraph: Graph convolutional network for skeleton-based gait recognition. In: IEEE international conference on image processing (ICIP) (pp. 2314–2318)
15. Li N, Zhao X (2022) A strong and robust skeleton-based gait recognition method with gait periodicity priors. IEEE Trans Multimed 25:3046–3058. https://doi.org/10.1109/TMM.2022.3154609
16. Lam TH, Cheung KH, Liu JN (2011) Gait flow image: a silhouette-based gait representation for human identification. Pattern Recognit 44(4):973–987. https://doi.org/10.1016/j.patcog.2010.10.011
17. Takemura N, Makihara Y, Muramatsu D, Echigo T, Yagi Y (2017) On input/output architectures for convolutional neural network-based cross-view gait recognition. IEEE Trans Circuits Syst Video Technol 29(9):2708–2719. https://doi.org/10.1109/TCSVT.2017.2760835
18. Elharrouss O, Almaadeed N, Al-Maadeed S, Bouridane A (2021) Gait recognition for person re-identification. J Supercomput 77:3653–3672. https://doi.org/10.1007/s11227-020-03409-5
19. Hou S, Cao C, Liu X, Huang Y (2020) Gait lateral network: Learning discriminative and compact representations for gait recognition, Springer International Publishing European conference on computer vision (pp. 382–398)
20. Wolf T, Babaee M, Rigoll G (2016) Multi-view gait recognition using 3D convolutional neural networks, In 2016 IEEE international conference on image processing (ICIP) (pp. 4165–4169)
21. Wang X, Yan K (2021) Gait classification through CNN-based ensemble learning. Multimed Tools Appl 80(1):1565–1581. https://doi.org/10.1007/s11042-020-09777-7
22. Wang X, Yan WQ (2020) Cross-view gait recognition through ensemble learning. Neural Comput Applic 32:7275–7287. https://doi.org/10.1007/s00521-019-04256-z
23. Mohammed A, Kora R (2023) A comprehensive review on ensemble deep learning: opportunities and challenges. J King Saud Univ - Comput Inf Sci 35(2):757–774. https://doi.org/10.1016/j.jksuci.2023.01.014
24. Rida I, Jiang X, Marcialis GL (2015) Human body part selection by group Lasso of motion for model-free gait recognition. IEEE Signal Process Lett 23(1):154–158. https://doi.org/10.1109/LSP.2015.2507200
25. Rokanujjaman M, Hossain MA, Islam MR (2012) Effective part selection for part-based gait identification, In 2012 7th IEEE International Conference on Electrical and Computer Engineering, (pp. 17–19)
26. Ghebleh A, Ebrahimi Moghaddam M (2018) Clothing-invariant human gait recognition using an adaptive outlier detection method. Multimed Tools Appl 77:8237–8257. https://doi.org/10.1007/s11042-017-4712-z
27. Yu S, Tan D, Tan T (2006) A framework for evaluating the effect of view angle, clothing and carrying condition on gait recognition, In Proc. 18th Int. Conf. Pattern Recog. (pp. 441–444)
28. Tan D, Huang K, Yu S, Tan T (2006) Efficient night gait recognition based on template matching, In 18th international conference on pattern recognition ICPR'06 (p. 3:1000–1003)
29. Song C, Huang Y, Huang Y, Jia N, Wang L (2019) Gaitnet: an end-to-end network for gait based human identification. Pattern Recognit 96:106988. https://doi.org/10.1016/j.patcog.2019.106988
30. Liu Z, Mao H, Wu CY, Feichtenhofer C, Darrell T, Xie S (2022) A convnet for the 2020s, In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 11976–11986)
31. Yaprak B, Gedikli E (2024) Different gait combinations based on multi-modal deep CNN architectures. Multimed. Tools Appl 83:83403–83425. https://doi.org/10.1007/s11042-024-18859-9
32. Zhang Y, Huang Y, Wang L, Yu S (2019) A comprehensive study on gait biometrics using a joint CNN-based method. Pattern Recognit 93:228–236. https://doi.org/10.1016/j.patcog.2019.04.023
33. Zhang Z, Tran L, Yin X, Atoum Y, Liu X, Wan J, Wang N (2019) Gait recognition via disentangled representation learning, In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4710–4719)
34. Teepe T, Gilg J, Herzog F, Hörmann S, Rigoll G (2022) Towards a deeper understanding of skeleton-based gait recognition, In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 1569–1577)
35. Gao S, Tan Z, Ning J, Hou B, Li L (2023) ResGait: gait feature refinement based on residual structure for gait recognition. Vis Comput 39(8):3455–3466. https://doi.org/10.1007/s00371-023-02973-0
36. Xu K, Jiang X, Sun T (2021) Gait recognition based on local graphical skeleton descriptor with pairwise similarity network. IEEE Trans Multimedia 24:3265–3275. https://doi.org/10.1109/TMM.2021.3095809
37. Meng C, He X, Tan Z, Luan L (2023) Gait recognition based on 3D human body reconstruction and multi-granular feature fusion. J Supercomput 79:12106–12125. https://doi.org/10.1007/s11227-023-05143-0
38. Usman SM, Khalid S, Bashir S (2021) A deep learning based ensemble learning method for epileptic Seizur prediction.

Comput Biol Med 136:104710. https://doi.org/10.1016/j.compbiomed.2021.104710

39. Kazmaier J, Van Vuuren JH (2022) The power of ensemble learning in sentiment analysis. Expert Syst Appl 187:115819. https://doi.org/10.1016/j.eswa.2021.115819
40. Ye H, Sun T, Xu K (2023) Gait recognition based on gait optical flow network with inherent feature pyramid. Appl Sci 13(19):10975. https://doi.org/10.3390/app131910975
41. Xiong J, Zou S, Tang J, Tjahjadi T (2024) MCDGait: multimodal co-learning distillation network with spatial-temporal graph reasoning for gait recognition in the wild. Vis Comput 40:7221–7234. https://doi.org/10.1007/s00371-024-03426-y
42. Takemura N, Makihara Y, Muramatsu D, Echigo T, Yagi Y (2018) IPSJ Trans Comput Vis Appl 10:1–14. https://doi.org/10.1186/s41074-018-0039-6. Multi-view large population gait dataset and its performance evaluation for cross-view gait recognition
43. Ekinci M, Gedikli E (2005) A novel approach on silhouette based human motion analysis for gait recognition. In International Symposium on Visual Computing (pp. 219–226)
44. Sagi O, Rokach L (2018) Ensemble learning: A survey. Wiley Interdiscip Rev Data Min Knowl Discov 8(4):e1249. https://doi.org/10.1002/widm.1249
45. Gupta A, Semwal VB (2020) Multiple task human gait analysis and identification: ensemble learning approach. In: Mohanty SN (ed) Emotion and information processing. Springer, Cham, pp 185–197. https://doi.org/10.1007/978-3-030-48849-9_12
46. Semwal VB, Gupta A, Lalwani P (2021) An optimized hybrid deep learning model using ensemble learning approach for human walking activities recognition. J Supercomput 77(11):12256–12279. https://doi.org/10.1007/s11227-021-03768-7
47. Ye M, Lan X, Leng Q, Shen J (2020) Cross-modality person re-identification via modality-aware collaborative ensemble learning. IEEE Trans Image Process 29:9387–9399. https://doi.org/10.1109/TIP.2020.2998275
48. Li Z, Shi Y, Ling H, Chen J, Wang Q, Zhou F (2022), June Reliability exploration with self-ensemble learning for domain adaptive person re-identification, In Proceedings of the AAAI conference on artificial intelligence, pp. 1527–1535
49. Liang D, Fan G, Lin G, Chen W, Pan X, Zhu H (2019) Three-stream convolutional neural network with multi-task and ensemble learning for 3d action recognition, In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops (pp. 0–0)
50. Xu Y, Cheng J, Wang L, Xia H, Liu F, Tao D (2018) Ensemble one-dimensional Convolution neural networks for skeleton-based action recognition. IEEE Signal Process Lett 25(7):1044–1048
51. Zhao L, Guo L, Zhang R, Xie X, Ye X (2022) MmGaitSet: multimodal based gait recognition for countering carrying and clothing changes. Appl Intell 52(2):2023–2036. https://doi.org/10.1007/s10489-021-02484-2
52. Li H, Qiu Y, Zhao H, Zhan J, Chen R, Wei T, Huang Z (2022) GaitSlice: A gait recognition model based on spatio-temporal slice features. Pattern Recognit 124:108453. https://doi.org/10.1016/j.patcog.2021.108453
53. Sepas-Moghaddam A, Etemad A (2020) View-invariant gait recognition with attentive recurrent learning of partial representations. IEEE Trans Biom Behav Identity Sci 3(1):124–137
54. Sepas-Moghaddam A, Ghorbani S, Troje NF, Etemad A (2021) Gait recognition using multi-scale partial representation transformation with capsules. In *2020* IEEE *25th international conference on pattern recognition (ICPR)* (pp. 8045–8052)
55. Ma K, Fu Y, Zheng D, Cao C, Hu X, Huang Y (2023) Dynamic aggregated network for gait recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 22076–22085)
56. Chen J, Wang Z, Zheng C, Zeng K, Zou Q, Cui L (2023) GaitAMR: Cross-view gait recognition via aggregated multi-feature representation. Inf Sci 636:118920. https://doi.org/10.1016/j.ins.2023.03.145
57. Wei T, Liu M, Zhao H, Li H (2024) Gmsn: an efficient multi-scale feature extraction network for gait recognition. Expert Syst Appl 252:124250. https://doi.org/10.1016/j.eswa.2024.124250
58. Pan X, Xie H, Zhang N, Li S (2024) GaitLRDF: gait recognition via local relevant feature representation and discriminative feature learning. Appl Intell 54(23):12476–12491. https://doi.org/10.1007/s10489-024-05837-9
59. Parashar A, Parashar A, Ding W, Shekhawat RS, Rida I (2023) Deep learning pipelines for recognition of gait biometrics with covariates: a comprehensive review. Artif Intell Rev 56(8):8889–8953. https://doi.org/10.1007/s10462-022-10365-4
60. Parashar A, Shekhawat RS, Ding W, Rida I (2022) Intra-class variations with deep learning-based gait analysis: A comprehensive survey of covariates and methods. Neurocomput 505:315–338. https://doi.org/10.1016/j.neucom.2022.07.002
61. Fan C, Liang J, Shen C, Hou S, Huang Y, Yu S (2023) Opengait: Revisiting gait recognition towards better practicality. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 9707–9716)
62. Wei S, Liu W, Wei F, Wang C, Xiong NN (2024) Gaitdlf: global and local fusion for skeleton-based gait recognition in the wild. J Supercomput 80(12):17606–17632. https://doi.org/10.1007/s11227-024-06089-7
63. Chen G, Chen X, Zheng C, Wang J, Liu X, Han Y (2024) Spatiotemporal smoothing aggregation enhanced multi-scale residual deep graph convolutional networks for skeleton-based gait recognition. Appl Intell 54(8):6154–6174. https://doi.org/10.1007/s10489-024-05422-0
64. Bilal M, Jianbiao H, Mushtaq H, Asim M, Ali G, ElAffendi M (2024) GaitSTAR: Spatial–Temporal Attention-Based Feature-Reweighting architecture for human gait recognition. Mathematics 12(16):2458